Support: VM/WSL Setup

Laboratórios de Sistemas e Serviços

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February 09, 2026

Introduction: Choose Your Environment

To ensure a consistent workspace, you must set up a Linux environment. Choose **one** of the following options based on your operating system:

1. **VirtualBox:** Best for Intel/AMD Windows and Linux PCs (Standard Choice).
2. **VMware Workstation:** High-performance alternative for Windows/Linux.
3. **UTM:** The **required** choice for Apple Silicon Macs (M1/M2/M3/M4 chips).
4. **WSL (Windows Subsystem for Linux):** Best for advanced Windows users who prefer terminal integration over a full GUI VM.

Part 1: Download the Class Disk

For **VirtualBox, VMware, and UTM**, you need the pre-configured Debian disk.

1. **Download:** Download Debian VM Disk (.vdi)
2. **Save:** Save to `Downloads` or `Documents`.

VM Credentials:

Field	Value
Username	`student`
Password	`password`

> **Format note:** This file is in `.vdi` format (VirtualBox native).

- **VirtualBox:** Use as is.
- **VMware/UTM:** You may need to convert it or install a fresh Debian ISO if the import fails (see Appendix A).

Option A: VirtualBox (Standard for Windows/Intel Mac)

Do not use this on Apple Silicon (M1/M2/M3/M4) Macs. Use Option C (UTM) instead.

1. Installation

1. **Download:** Go to virtualbox.org/wiki/Downloads.
2. **Install App:** Download and run the installer for your OS ("Windows hosts" or "macOS / Intel hosts").
3. **Install Extension Pack:** Download the "Oracle VM VirtualBox Extension Pack" from the same page and double-click it to install.

2. Create VM

1. Open VirtualBox, click **New**.
   - **Name:** `Debian LSS`
   - **Type:** `Linux` / **Version:** `Debian (64-bit)`
2. **RAM:** 4096 MB (4 GB) recommended.
3. **Hard Disk:** Select **"Use an Existing Virtual Hard Disk File"**.
4. Click the folder icon, navigate to the `.vdi` file you downloaded, and select it.
5. Click **Create**.

3. Drivers (Guest Additions)

Guest Additions enable clipboard sharing, drag-and-drop, and automatic screen resizing.

1. Start the VM and log in with the credentials above.
2. In the VM menu: **Devices > Insert Guest Additions CD image**.
3. Open a terminal in the VM and run:

```
sudo apt update
sudo apt install build-essential dkms linux-headers-$(uname -r)
sudo mkdir -p /mnt/cdrom
sudo mount /dev/cdrom /mnt/cdrom
```

```
sudo /mnt/cdrom/VBoxLinuxAdditions.run
sudo reboot
```

Alternative method (simpler, uses Debian packages):

```
sudo apt update
sudo apt install virtualbox-guest-utils virtualbox-guest-x11
sudo reboot
```

4. Verify

After reboot, log in and open a terminal:

```
uname -a
```

You should see a Linux kernel version. Try resizing the VM window; the guest desktop should resize automatically.

Option B: VMware Workstation (Windows/Linux)

VMware Workstation Pro is now **free for personal use** and often provides better performance than VirtualBox.

1. Installation

1. **Download:** Create a Broadcom account and download **VMware Workstation Pro** (Windows) or **VMware Fusion** (Mac – Intel only).
2. **Install:** Run the installer. Select "I want to license for Personal Use" (no key required).

2. Create VM (Importing VDI)

VMware uses `.vmdk` format natively. You can try selecting the `.vdi` by choosing "All Files", but conversion is safer (see Appendix A).

1. Click **"Create a New Virtual Machine"** > **Custom (Advanced)**.
2. **Guest OS:** Linux > Debian 12 (64-bit).
3. **Disk:** Choose **"Use an existing virtual disk"**.
4. Browse for your converted `.vmdk` file (or try the `.vdi` directly).
5. When asked to **"Convert"** the format, click **Yes**.

3. Drivers (Open-VM-Tools)

VMware uses open-source drivers that enable clipboard sharing, drag-and-drop, and automatic screen resizing.

1. Start the VM and log in with the credentials above.
2. Open a terminal and run:

```
sudo apt update
sudo apt install open-vm-tools-desktop
sudo reboot
```

4. Verify

After reboot, log in and open a terminal:

```
uname -a
```

Try copying text between the host and guest to confirm clipboard integration works.

Option C: UTM (Apple Silicon Mac M1/M2/M3/M4)

This is the only performant option for modern Macs.

1. Installation

1. Download **UTM** from mac.getutm.app (Free) or the Mac App Store (Paid/Donation).
2. **Important:** The class disk is `.vdi` (x86 architecture). Emulating x86 on Apple Silicon is **slow**.
   - *Recommendation:* It is highly recommended to **install a fresh ARM64 Debian** instead of using the class disk.
   - *If you must use the class disk:* You need to convert it to `.qcow2` first (see Appendix A) and accept slow emulation speeds.

2. Create VM (Fresh Install Method – Recommended)

1. Download the **Debian ARM64 ISO** from debian.org.
2. Open UTM > **Create a New Virtual Machine**.
3. **Virtualize** (Fast) > **Linux**.
4. **Boot Image:** Select the Debian ARM64 ISO.
5. Follow the installer steps. Use the same credentials as the class disk for consistency.

3. Drivers (SPICE Agent)

To get clipboard sharing and dynamic resolution:

1. Open a terminal in the VM.

2. Run:

```
sudo apt update
sudo apt install spice-vdagent
sudo reboot
```

4. Verify

After reboot, log in and open a terminal:

```
uname -a
```

You should see `aarch64` in the output, confirming the ARM64 architecture.

Option D: Windows Subsystem for Linux (WSL)

Runs Linux natively alongside Windows. High performance, but no virtual desktop window by default.

1. Installation

1. Open **PowerShell** as Administrator.
2. Run:

```
wsl --install
```

3. **Reboot** your computer.
4. After reboot, "Ubuntu" will open. Create a username and password.

2. Keep WSL Updated

After initial setup, ensure WSL and the distribution are up to date:

```
wsl --update
```

3. Optimize Networking (Mirrored Mode)

To make WSL behave like a real PC on your network (fixing many connectivity issues):

1. In Windows, press `Win+R`, type `%UserProfile%`, and hit Enter.
2. Create a file named `.wslconfig` (make sure it is not `.wslconfig.txt`).
3. Paste this content:

```
[wsl2]
networkingMode=mirrored
```

```
dnsTunneling=true
autoProxy=true
```

4. Restart WSL from PowerShell: `wsl --shutdown`.

4. GUI Apps

WSL supports GUI applications out of the box (via WSLg). Test it:

```
sudo apt update && sudo apt install mousepad thunar
mousepad
```

A text editor window should appear on your Windows desktop.

5. Verify

```
cat /etc/os-release
uname -r
```

You should see the Ubuntu version and a Linux kernel version containing `WSL` or `microsoft`.

Post-Setup: Common First Steps

Regardless of which option you chose, perform the following steps in your new Linux environment.

1. Update the System

```
sudo apt update && sudo apt upgrade -y
```

2. Install Essential Tools

```
sudo apt install -y git curl wget vim nano net-tools
```

3. Configure Git

```
git config --global user.name "Your Name"
git config --global user.email "your.email@ua.pt"
```

4. Verify Everything Works

```
echo "Hello from Linux!"
uname -a
git --version
```

Troubleshooting

VT-x / AMD-V Not Enabled

Symptom: VirtualBox or VMware fails to start the VM with a “VT-x is disabled” or “AMD-V is not available” error.

Fix: Restart your computer, enter the BIOS/UEFI settings (usually by pressing `F2`, `F12`, `Del`, or `Esc` during boot), and enable **Intel VT-x** or **AMD-V** (sometimes labeled “SVM Mode”) under the CPU or Security settings.

Hyper-V Conflicts on Windows

Symptom: VirtualBox reports that Hyper-V is active and VT-x cannot be used.

Fix: Hyper-V and VirtualBox compete for hardware virtualization. You can disable Hyper-V:

```
bcdedit /set hypervisorlaunchtype off
```

Reboot after running this command. To re-enable Hyper-V later:

```
bcdedit /set hypervisorlaunchtype auto
```

> **Note:** Disabling Hyper-V also disables WSL 2, Windows Sandbox, and Credential Guard. If you need those, consider using VMware or WSL instead of VirtualBox.

Network Not Working in VM

Symptom: The VM has no internet access.

Fix: Check the VM network adapter settings:

- Ensure the adapter is set to **NAT** (the simplest mode).
- Verify your host machine has internet access.
- Inside the VM, check the adapter status: `ip addr show` and `ping -c 3 8.8.8.8`.
- Restart networking: `sudo systemctl restart NetworkManager`.

Secure Boot Issues

Symptom: The VM fails to boot or kernel modules (like VirtualBox Guest Additions) fail to load.

Fix: Some systems require Secure Boot to be disabled in BIOS/UEFI for third-party kernel modules to load. Disable it in the BIOS/UEFI settings under the Security or Boot tab.

Appendix A: Converting the Disk (.vdi)

If you are not using VirtualBox, you may need to convert the downloaded `.vdi` file.

For VMware (VDI to VMDK)

Requires VirtualBox to be installed (for VBoxManage). Run in Command Prompt:

```
"C:\Program Files\Oracle\VirtualBox\VBoxManage.exe" ^
  clonemedium disk "C:\Path\To\Input.vdi" ^
  "C:\Path\To\Output.vmdk" --format VMDK
```

For UTM (VDI to QCOW2)

Requires `qemu-img`. Install via Homebrew on macOS:

```
brew install qemu
qemu-img convert -f vdi -O qcow2 Input.vdi Output.qcow2
```

On Windows, `qemu-img` is available by installing QEMU for Windows, or you can run the conversion inside WSL:

```
sudo apt install qemu-utils
qemu-img convert -f vdi -O qcow2 /mnt/c/Path/To/Input.vdi Output.qcow2
```